

Econ 131
Spring 2026
Danny Yagan

Problem Set 2 — SOLUTIONS

1. True/False/Uncertain with Explanation [25 points]

Determine whether each statement is true, false, or uncertain and explain why. Answers with no explanation will receive no points. [5 points each]

- (a) A dividend income tax cannot raise revenue without reducing investment.

False. In the “new view” of dividend taxation, the tax can fall mainly on payouts from existing capital and need not change the marginal incentive to invest out of retained earnings. In that case, the government can raise revenue with little or no effect on current investment. In other settings, such as new-equity-financed investment, dividend taxation can reduce investment.

- (b) If the income distribution in the top tax bracket becomes thinner (individuals above the bracket threshold become more concentrated at incomes near the threshold), the government should increase the top marginal tax rate.

False. A thinner top tail means there is less income far above the threshold, so the mechanical revenue gain from raising the top rate is smaller. In the standard top-rate formula,

$$\frac{\tau^*}{1 - \tau^*} = \frac{1 - \bar{g}}{ae},$$

a thinner top tail corresponds to a larger Pareto parameter a , which lowers the optimal top tax rate holding $\bar{g}$ and e fixed.

- (c) If leisure is a normal good, then an increase in non-labor income creates a negative income effect on labor supply.

True. If leisure is normal, higher non-labor income increases desired leisure. Since labor supply equals time endowment minus leisure, labor supply falls. Thus the income effect on labor supply is negative.

- (d) A lump-sum tax in period 1 that is independent of savings changes the intertemporal relative price between current and future consumption.

False. A lump-sum tax changes lifetime resources, but not the relative price of current versus future consumption. The intertemporal price is determined by the after-tax return to saving, not by a lump-sum tax that is independent of savings decisions.

- (e) The Pareto parameter for the entire income distribution is $1/2$.

False. Using the class definition

$$a = \frac{\bar{z}}{\bar{z} - z^*},$$

for the entire distribution we set $z^* = 0$, so

$$a = \frac{\bar{z}}{\bar{z}} = 1.$$

So the Pareto parameter for the whole distribution is 1 under this definition, not 1/2.

2. Business Taxation and Investment [26 points]

Consider a firm that chooses capital K to maximize profit. Let $F(K)$ denote the firm's gross profit from using capital, where $F'(K) > 0$ and $F''(K) < 0$. Let r denote the per-unit cost of capital.

Throughout, assume the firm chooses K optimally and interior solutions exist.

- (a) [4 points] Suppose there is **no tax**. The firm solves

$$\max_K F(K) - rK.$$

Write the first-order condition for the firm's optimal choice of capital.

The first-order condition is

$$F'(K) - r = 0,$$

so the optimal capital stock satisfies

$$F'(K^*) = r.$$

The firm invests until the marginal profit from capital equals the marginal cost of capital.

- (b) [5 points] Now suppose the government imposes a **business income tax** at rate τ on **gross profit**, so the firm solves

$$\max_K (1 - \tau)F(K) - rK.$$

Derive the first-order condition. Does this tax increase, decrease, or leave unchanged the firm's desired level of investment? Briefly explain.

The first-order condition is

$$(1 - \tau)F'(K) - r = 0,$$

or

$$F'(K) = \frac{r}{1 - \tau}.$$

Since $\tau > 0$ implies $\frac{r}{1 - \tau} > r$, the firm now requires a higher marginal product of capital to invest. Because $F''(K) < 0$, this means the firm chooses a lower level of capital. Thus the tax reduces investment.

- (c) [5 points] Instead suppose the business income tax applies to profits **after deducting capital costs**, so the firm solves

$$\max_K (1 - \tau)[F(K) - rK].$$

Derive the first-order condition. Does this tax affect the firm's desired investment? Briefly explain.

The first-order condition is

$$(1 - \tau)[F'(K) - r] = 0.$$

Since $1 - \tau > 0$, this simplifies to

$$F'(K) = r.$$

This is exactly the same condition as in part (a), so the firm's desired investment is unchanged. In this simple model, the tax is neutral because the marginal return and marginal cost are both reduced proportionally. In other words, tax falls on pure profits.

- (d) [4 points] In 2–3 sentences, explain the economic intuition for why the answers in parts (b) and (c) are different.

In part (b), the tax reduces the marginal return to capital, but the firm still pays the full marginal cost of capital. This raises the effective user cost of capital and discourages investment. In part (c), the tax applies only to net profits after deducting capital costs, so both the marginal return and marginal cost are scaled in the same way. The investment condition is unchanged, so the tax is neutral in this model.

- (e) [4 points] Historically, many capital costs have not been deductible immediately from taxable profits. Explain why this makes the tax system less like the case in part (c) and more like the case in part (b).

If capital costs are deducted only gradually over time, then their present discounted value is less than a full immediate deduction. This means the tax system does not fully offset the marginal cost of investment, so it is less like part (c) and more like part (b). As a result, the tax system tends to discourage investment.

- (f) [4 points] Suppose the tax system is the one from part (b). Now suppose that the government offers an investment tax credit c : for every dollar spent on capital, the firm gets to reduce its tax bill by $\$c$. What value does c need to be in order for the new tax system as a whole to not distort the firm's choice of capital relative to (a)?

Under part (b), suppose the government offers an investment tax credit at rate c per dollar of capital spending. Then the firm's problem becomes

$$\max_K (1 - \tau)F(K) - rK + crK = \max_K (1 - \tau)F(K) - (1 - c)rK.$$

The first-order condition is

$$(1 - \tau)F'(K) - (1 - c)r = 0,$$

so

$$F'(K) = \frac{(1 - c)r}{1 - \tau}.$$

To eliminate the distortion and match the no-tax benchmark from part (a), we want

$$F'(K) = r.$$

So we need

$$\frac{1 - c}{1 - \tau} = 1.$$

Solving gives

$$c = \tau.$$

Therefore, the investment tax credit must equal the business tax rate in order to fully offset the distortion in this simple setup.

3. Labor Income Taxes and Labor Supply [18 points]

An individual chooses labor supply $n \geq 0$. Consumption is

$$c = (1 - \tau)wn + R,$$

where w is the pre-tax wage, τ is the labor income tax rate, and R is non-labor income.

Consider the two utility functions

$$\begin{aligned} \text{(I)} \quad u_1(c, n) &= c - \frac{n^{1+1/\varepsilon}}{1 + 1/\varepsilon}, \\ \text{(II)} \quad u_2(c, n) &= \log(c) - \alpha \frac{n^{1+1/\varepsilon}}{1 + 1/\varepsilon}, \end{aligned}$$

where $\varepsilon > 0$ and $\alpha > 0$.

- (a) [3 points] For case (I), derive the first-order condition for optimal labor supply. Show that labor supply depends on the net-of-tax wage $(1 - \tau)w$.

For case (I),

$$u_1(c, n) = c - \frac{n^{1+1/\varepsilon}}{1 + 1/\varepsilon}, \quad c = (1 - \tau)wn + R.$$

Substituting for c gives

$$u_1 = (1 - \tau)wn + R - \frac{n^{1+1/\varepsilon}}{1 + 1/\varepsilon}.$$

The first-order condition with respect to n is

$$(1 - \tau)w - n^{1/\varepsilon} = 0.$$

So optimal labor supply satisfies

$$n^{1/\varepsilon} = (1 - \tau)w, \quad \text{or} \quad n = [(1 - \tau)w]^\varepsilon.$$

Thus labor supply depends on the net-of-tax wage $(1 - \tau)w$.

- (b) [3 points] In case (I), if the tax rate τ rises, is there an income effect on labor supply? Is there a substitution effect? What happens to labor supply overall?

In case (I), there is no income effect on labor supply because utility is quasi-linear in consumption. A higher tax rate τ lowers the net-of-tax wage $(1 - \tau)w$, so the return to working falls. This creates a substitution effect away from labor. Therefore labor supply falls overall.

- (c) [3 points] In case (II), if the tax rate τ rises, explain why labor supply is affected by both an income effect and a substitution effect. Do these two effects go in the same direction or opposite directions?

In case (II),

$$u_2(c, n) = \log(c) - \alpha \frac{n^{1+1/\varepsilon}}{1 + 1/\varepsilon}, \quad c = (1 - \tau)wn + R.$$

A higher labor income tax lowers the net wage, so the reward to working falls. This creates a substitution effect that lowers labor supply. At the same time, the tax reduces consumption for any given level of labor supply, so the individual is poorer. Because consumption enters utility nonlinearly through $\log(c)$, this creates an income effect. The income effect tends to raise labor supply, since the individual may work more when poorer. Thus the substitution effect lowers labor supply, while the income effect raises labor supply. The two effects go in opposite directions.

- (d) [3 points] Which case would generate a larger gap between the compensated and uncompensated labor-supply elasticities? Explain briefly.

Case (II) would generate the larger gap. The compensated elasticity isolates the substitution effect, while the uncompensated elasticity includes both substitution and income effects. In case (II), income effects are present, so the two elasticities differ more. In case (I), income effects are absent, so compensated and uncompensated elasticities are the same in this benchmark.

- (e) [3 points] Suppose the government increases τ but also adjusts R so that the individual can still reach the same utility level as before. Is this change closer to an uncompensated or a compensated tax change? In this situation, which effect on labor supply is being isolated?

This is a compensated tax change, because non-labor income R is adjusted so that the individual remains at the same utility level after the tax change. Holding utility fixed removes the income effect, so the remaining behavioral response isolates the substitution effect.

- (f) [3 points] In empirical public finance, labor-supply responses are often summarized using the elasticity of labor income with respect to the net-of-tax rate, $1 - \tau$. Based on your answers above, would you expect this elasticity to be larger in absolute value under case (I) or case (II)? Explain briefly.

We would generally expect this elasticity to be larger in absolute value under case (I). In case (I), there is no income effect, so the response to a change in the net-of-tax rate reflects only the substitution effect. In case (II), the income effect works in the opposite direction and partly offsets the substitution effect, so the observed uncompensated elasticity is typically smaller in absolute value.

End of Solutions